



clock movement = $0.12 \times 1.71 = 0.2\text{mW}$

Energy consumed during 24 hours = $0.2\text{mW} \times 24 = 4.8\text{mWh}$

Energy stored in the capacitor = $0.5 \times C1 \times V \times V = 0.5 \times 100 \times 1.99 \times 1.99 = 198 \text{ joules}$

Energy stored = $(198/3600) \times$

PARTS LIST

Semiconductors:

IC1 - TL431 shunt regulator IC
D1-D2 - 1N5819 Schottky diode
D3-D5 - 1N4007 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon), unless stated otherwise:

R1 - 330-ohm, 1-watt
R2 - 220-ohm

Capacitors:

C1 - 100F, 2.7V super capcitor
C2 - 4700 μ F, 10V electrolytic

Miscellaneous:

PV1, PV2 - 6V, 100mA solar panel
- Small piece of general-purpose PCB for mounting the resistors and capacitors

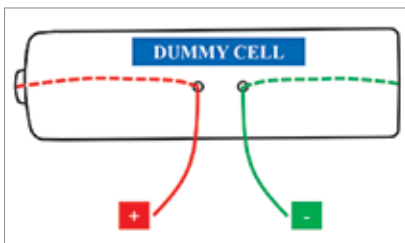


Fig. 6: Dummy cell with interconnecting wires for quickly connecting to the clock

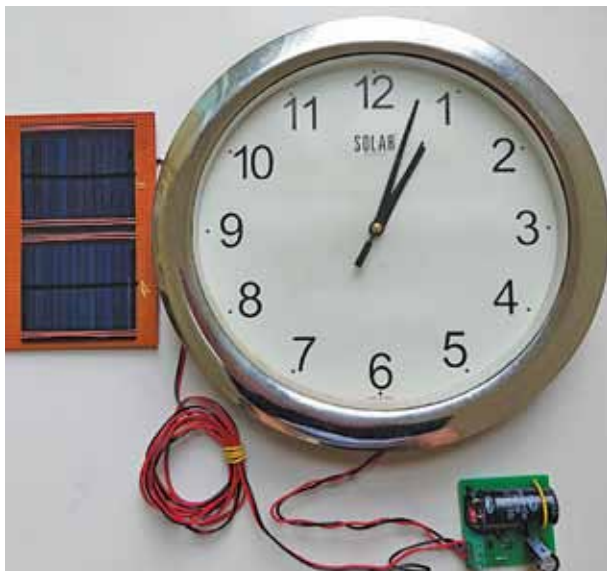


Fig. 7: PV panels and clock interconnections

$1000 = 55\text{mWh}$

From above calculations it is seen that, we have almost 11 times energy available in the capacitors on per day basis (under ideal conditions). However, there are self-discharge losses in the capacitor and losses in R2 and diodes as well.

Considering all the losses we can safely assume that the stored power in the capacitor will be sufficient for two to three days. This factor helps in non-stop running of clock on rainy days when the sunlight is very less.

Fig. 5 shows the filtered voltage waveform +Vclock. With capacitor C2 connected at the output, the pulse peak voltage drops to less than 200mV. Also, the value of resistance has been increased to 220 Ω to reduce current through the diodes. These waveforms were captured when the C1 voltage was 1.08V. Thus, it is clear that the clock works down to 1V.

Fig. 6 shows proposed AA size dummy cell made out of plastic material. It has metallic positive and negative terminals. Two wires are connected to these terminals and brought out through the holes provided. These wires are connected to +Vclock and -Vclock. User can simply insert this dummy cell in the clock moment.

Wall clocks and desk clocks are inside the buildings, hence they do not receive direct sunlight. Therefore, two PV panels have been used so that the circuit works on the diffused ambient light available inside the rooms. There is no need to mount the panels in direct sunlight.

The panels can be mounted on the same wall on which the clock is present. We have to ensure that throughout the day, the panels get good ambient light. If this light is not sufficient then we can mount the panels near the windows where light intensity is higher. Two PV panels also ensure that during rainy season the circuit works properly.

Now replace the dummy cell with the prototype shown in Fig. 7.

If there is some place inside the room where direct sunlight is available for a few hours, it may be possible to remove one PV panel. With direct sunlight, a single panel can fully charge the capacitor quickly. This will help in reducing the cost to some extent, but it will need long-term testing in all seasons to ensure nonstop operation.

The above design demonstrates use of supercapacitors for storing small amount of energy. As tens of billions of clocks are being used in a country like India, it offers big potential for reducing the use of chemical batteries and thus reducing hazardous waste. Also, it will generate demand for tens of billions of supercapacitors, which will bring down the cost.

There are many other areas where these capacitors could be used for energy storage and reduce the country's dependence on imported lithium and other costly metals. As each house will generate a small amount of solar power, its cumulative effect will be significant. Need of the hour is to identify many more such applications, which can contribute towards making an Atmanirbhar and Swachh Bharat. **EFY**

Dr Vijay Deshpande has worked for more than 30 years as electronics hardware engineer in several companies and recently retired from Honeywell India. He now likes to work on solar energy projects.

